The University of Texas at Austin
Practice Materials for In-Term Exam 2

M339G Predictive Analytics

November 13, 2026

Instructions: This is a closed book and closed notes exam. The maximal score on the exam is 100 points.
Time: 50 minutes

CONCEPTUAL QUESTIONS

Problem 2.1. (10 points) What is the crucial difference in the assumptions between LDA and QDA? What consequence does that change have on the classification criterion?

Problem 2.2. (10 points) Explain how naive Bayes is different from LDA. Provide one advantage which stems from this different approach. Are there disadvantages?

Problem 2.3. (10 points) Describe the differences and the similarities between LDA and the logistic regression for a two-class classification problem.

Problem 2.4. (10 points) Justify this statement from the textbook:

"Neither of QDA nor naive Bayes is a special case the other."

FREE RESPONSE PROBLEMS

Problem 2.5. (10 points) *Source: Pitman's "Probability."*
Heights and weights of a large group of people follow a bivariate normal distribution, with correlation 0.75. Of the people **at** the 90th percentile of weights, about what percentage are **above** the 90th percentile of heights?

Problem 2.6. (15 points) *Source: An old SRM manual.*
For a regression tree, two nodes in the tree - defining regions R_1 and R_2 - have the following values of the response variable (on the training set):

$$R_1 : 2, 3, 3, 4, 5$$
$$R_2 : 2, 4, 6$$

The tree is pruned using cost complexity pruning. The split creating R_1 and R_2 is the optimal one to prune. Determine the smallest value of α for which this pruning will occur.

Problem 2.7. (20 points) A classification tree is being constructed to predict whether an insured will file a claim. A random sample of 200 insureds contains 60 who filed a claim. You are considering two splits:

- Split A: One node has 50 observations with 30 claims and one node has 150 observations with 30 claims.
- Split B: One node has 70 observations with 40 claims and one node has 130 observations with 20 claims.

The total Gini index after a split is the weighted average of the Gini index at each node, with the weights proportional to the number of observations in each node. The total entropy after a split is the weighted average of the entropy at each node, with the weights proportional to the number of observations in each node.

Determine which of the following statements is/are true.

I. Split A is preferred based on the total Gini index.
II. Split B is preferred based on the total entropy.
III. Splits A and B result in the same number of classification errors.

a. I only
b. II only
c. I and III only
d. II and III only
e. The correct answer is not given by a., b., c., or d.

Problem 2.8. (20 points) Consider the following observations of (X, Y) with X being the predictor and Y being the response:

$$(1, 4), \quad (2, 6), \quad (3, 5), \quad (6, 1).$$

After one iteration of recursive binary splitting, there are two groups of observations. Find the members of the two groups.

Problem 2.9. (20 points) Consider the following observations of (X, Y) with X being the predictor and Y being the response:

$$(0, 8), \quad (2, 6), \quad (4, 5), \quad (6, 7).$$

After one iteration of recursive binary splitting, there are two groups of observations. Find the members of the two groups.

MULTIPLE CHOICE QUESTIONS

Problem 2.10. (5 points) *Source: SRM Sample Problem #29.*
Determine which of the following considerations may make decision trees preferable to other statistical methods.

I. Decision trees are easily interpretable.
II. Decision trees can be displayed graphically.
III. Decision trees are easier to explain than linear regression models.

a. None.
b. I and II only.
c. I and III only.
d. II and III only.
e. The correct answer is not given above.

Problem 2.11. (5 points) Determine which of the following statements about K-means clustering are true.

I. Each iteration of the K-means algorithm can never increase the total within-cluster variation.
II. The K-means algorithm is guaranteed to find the partition of the observations which globally minimizes the total within-cluster variation.
III. The clusters obtained can change if the variables are standardized before running the algorithm.

a. None.
b. I and II only.
c. I and III only.
d. II and III only.
e. The correct answer is not given above.

Problem 2.12. (5 points) Determine which of the following statements about choosing the number of clusters K in K-means clustering are true.

I. Choosing K to minimize the total within-cluster variation is not a sensible criterion.
II. One sensible heuristic for choosing K would be to plot the total within-cluster variation against K and pick the value of K after which further increases give only small reductions.
III. The clusters obtained with $K = 3$ can always be obtained by splitting one of the clusters obtained with $K = 2$.

a. None.
b. I and II only.
c. I and III only.
d. II and III only.
e. The correct answer is not given above.

Problem 2.13. (5 points) Determine which of the following statements about classification trees are true.

I. When growing a classification tree, the classification error rate is preferred over the Gini index as the splitting criterion because it is more sensitive to node purity.
II. A split may produce two terminal nodes which have the same predicted class.
III. The Gini index of a node is small when most of the observations in that node belong to a single class.

a. None.
b. I and II only.
c. I and III only.
d. II and III only.
e. The correct answer is not given above.

The above problems are **in addition** to your past homework assignments and in-class work. Do not forget to re-solve those!